CLASS SETS AND EISENSTEIN CONGRUENCES ON COMPACT EXCEPTIONAL GROUPS

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ABSTRACT. Let G be a reductive group over $\mathbb{Q}$ with $G(\mathbb{R})$ compact and let $K \subset G(\mathbb{A}_f)$ be compact open. A theorem of Martin and Wakatsuki, extending Martin's work on definite quaternion algebras, says that if a prime ℓ divides the mass $\sum_x 1/|\Gamma_x|$ of the class set $G(\mathbb{Q})\backslash G(\mathbb{A}_f)/K$ then some cuspidal eigenform of level K is Hecke congruent to the constant function modulo ℓ. We record the form-level version of this statement—the primes at which an element of the cuspidal Eisenstein component is congruent to the constant function are *exactly* the primes of the mass numerator—whose elementary core we have verified in Lean 4, and we use it as an instrument. By Gross's mass formula this identifies the Eisenstein primes as the primes of the Bernoulli numbers B_{d_i} at the Weyl degrees d_i of G and of the cyclotomic factors $p^{d_i}-1$ at the level primes. We test the theorem by computing class sets and Hecke operators that were previously out of reach: the compact G_2 at Iwahori level p for $p = 7, 11, 13, 17$ (class numbers $29, 187, 523, 2280$), and the compact F_4 at level one and at the parahoric level Q_3 (15 classes; two commuting 15×15 Hecke matrices at 2). At every level the observed congruence primes are exactly the predicted ones, and the carriers are determined: on F_4 the primes 691 and 73 are the Eisenstein congruences of Δ and of the newform in $S_{12}(\Gamma_0(3))$, transported through the theta correspondence, while 41 is carried by a form invisible to every classical theta series of weight 12. For the compact E_8, whose class set has never been enumerated, the theorem proves Eisenstein congruences at the primes of the numerators of $B_{12}, B_{18}, B_{20}, B_{24}, B_{30}$. On the Niemeier genus the theorem recovers, from the mass alone, the congruence primes $131, 283, 593, 617, 691, 3617, 43867$ found in the published Hecke data of Chenevier and Lannes. The identification of the remaining F_4 eigen-systems, including three stable forms on smaller groups, is the subject of Part II.

1. INTRODUCTION

Gross [12] observed that when $G(\mathbb{R})$ is compact, automorphic forms on G of trivial weight are simply functions on the finite set $X_K = G(\mathbb{Q})\backslash G(\mathbb{A}_f)/K$, and Hecke operators are integer matrices counting adjacencies between classes. The constant function is always an eigenvector; its eigen-system is the analogue of the Eisenstein series. Following Mazur's study of the Eisenstein ideal for GL_2 [23], and Ramanujan's congruence $\tau(n) \equiv \sigma_{11}(n) \pmod{691}$ which is its prototype, one asks at which primes a cuspidal eigenform on G can be congruent to the constant function.

Martin [21] answered this for definite quaternion algebras and Martin–Wakatsuki [22] for all reductive G compact at infinity: if ℓ divides the mass $M = \sum_x 1/|\Gamma_x|$, there is a cuspidal eigenform Hecke congruent to the constant function modulo ℓ (their Theorem 2.1; the depth of the congruence is at least $v_\ell(M)$). We were unaware of this work when the computations below were begun and rediscovered the argument; the statement we use is the following form-level version, which is essentially their Theorem 2.1 together with their Lemma 3.4.

Theorem 1.1 (after Martin–Wakatsuki)**.** *Let $L = \operatorname{lcm} |\Gamma_x|$, $e_x = L/|\Gamma_x|$, $g = \gcd_x e_x$ and $E = LM/g \in \mathbb{Z}$. For a prime ℓ the following are equivalent:*

(i) *$\ell \mid E$;*

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(ii) *there is an integer-valued cuspidal function f on X_K (i.e. $\sum_x e_x f(x) = 0$) with $f \equiv 1 \pmod{\ell}$;*
(iii) *there is an element f of the cuspidal part of the Eisenstein component of $\mathbb{Z}_\ell^{X_K}$ with $f \equiv c \cdot 1 \pmod{\lambda}$ for a unit c.*

When they hold there is a cuspidal Hecke eigenform whose eigenvalues are all congruent to those of the constant function modulo a prime above ℓ.

The integer E is the numerator of M up to primes dividing the $|\Gamma_x|$; in every example below it equals the numerator of M in lowest terms, with one instructive exception. The proof uses only two properties of the Hecke matrices, which are also the checks one runs on any computed Brandt matrix: the constant function is an eigenvector, and the matrices are symmetric for the mass-weighted inner product. What is new here is not the theorem but its use: the proof has been formalised and checked in Lean 4 against Mathlib (Appendix B), and the statement is tested, at the form level and with the carriers identified, on the compact exceptional groups, where it had not been applied.

Gross's mass formula [10] makes Theorem 1.1 predictive. For G of rank r with Weyl degrees $d_1, \dots, d_r$, split at all finite places, with K hyperspecial away from a finite set S and Iwahori (or a fixed parahoric) at $p \in S$,

$$M(K_S) = 2^{-r} \prod_{i=1}^{r} |\zeta(1 - d_i)| \cdot \prod_{p \in S} \prod_{i=1}^{r} \frac{p^{d_i} - 1}{p - 1} \cdot c_S, \tag{1}$$

where c_S is the index of the parahoric in the Iwahori subgroup. The Eisenstein primes therefore come in two kinds: *Bernoulli-type* primes dividing the numerators of $\zeta(1 - d_i) = -B_{d_i}/d_i$, and *cyclotomic-type* primes dividing $p^{d_i} - 1$ for a level prime p; a prime ℓ is of the second kind at level p exactly when $\mathrm{ord}_\ell(p)$ divides some Weyl degree. For G_2 ($d = 2, 6$) the level-one mass is $1/12096$ and there is no Bernoulli prime; for F_4 ($d = 2, 6, 8, 12$) the level-one mass is $691/380414361600$ and 691 is Ramanujan's prime; for E_8 ($d = 2, 8, 12, 14, 18, 20, 24, 30$) the numerator is the product of the numerators of $B_{12}, B_{18}, B_{20}, B_{24}, B_{30}$.

The second part of the paper tests the theorem on data. We compute the class sets and several Hecke operators of the compact G_2 at Iwahori level p for $p = 7, 11, 13, 17$, extending the computations of Lansky and Pollack [15] (levels ≤ 7); the class numbers 187, 523 and 2280 are new, and the mass is recovered exactly at each level. We compute the class set of the compact F_4 at the parahoric level Q_3 associated with the first fundamental coweight (nine classes on the standard Albert lattice, six on the Elkies–Gross lattice), and two commuting Hecke operators at 2 on it. At level one we recover, from lattice enumeration alone, the eigenvalues predicted by the theta lift of Δ [25, 26]; at level Q_3 we determine the carriers of the three Eisenstein primes, and leave the identification of the remaining eigen-systems—which turns out to involve compact G_2-forms of weight $2\omega_2$, an SO(7)-form on the genus of E_7, and a stable genus-2 parameter—to Part II [1]. In every case the primes at which a cuspidal eigenform is congruent to the constant function are exactly the primes of E. Finally, we run the same test on the genus of even unimodular lattices of rank 24, using the neighbour counts published by Chenevier and Lannes [6]: the seven primes of the mass numerator are exactly the primes carrying form-level Eisenstein congruences, each Bernoulli numerator on the eigenform theory would assign it; the weights of the Hecke inner product on this genus are the reciprocals of the Weyl groups times the umbral groups of Cheng–Duncan–Harvey, and two of the Eisenstein eigenforms are explicit polynomials in the Coxeter number.

The data also show *which* eigenforms carry the Eisenstein primes. At G_2 level p every cyclotomic prime $\ell \mid (p^2 \mp p + 1)$ is carried by a non-tempered Arthur-type lift of a weight-6 newform f

of level p, and the G_2 congruence is the classical congruence $f \equiv E_6 \pmod{\ell}$ transported (Theorem 4.1); the primes from $p \mp 1$ are carried by tempered forms. This is the same mechanism that explains 691, 73 and 41 on F_4.

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2. The existence theorem

2.1. **Setting.** Let G be reductive over $\mathbb{Q}$ with $G(\mathbb{R})$ compact, $K \subset G(\mathbb{A}_f)$ compact open, and $X = X_K = \{x_1, \dots, x_h\}$ the class set, with finite stabilisers Γ_i and weights $w_i = 1/|\Gamma_i|$. Integer-valued forms of trivial weight are $\mathbb{Z}^X$. A Hecke operator T (for a double coset KgK) acts on $\mathbb{Z}^X$ by an integer matrix B with

(H1) $B\mathbf{1} = N\mathbf{1}$ for an integer N (the number of K-neighbours);
(H2) $w_i B_{ij} = w_j B_{ji}$.

By (H2) the pairing $\langle f, g\rangle = \sum_i w_i f_i g_i$ is $\mathbb{T}$-invariant, so the Hecke algebra $\mathbb{T} \subset \mathrm{End}(\mathbb{Z}^X)$ is a commutative ring of self-adjoint operators. Let $L = \operatorname{lcm}|\Gamma_i|$, $e_i = Lw_i \in \mathbb{Z}$, $g = \gcd e_i$, and

$$E := \frac{1}{g}\sum_i e_i = \frac{LM}{g} \in \mathbb{Z}.$$

The *cuspidal lattice* is $C = \{f \in \mathbb{Z}^X : \sum_i e_i f_i = 0\} = \mathbb{Z}^X \cap \mathbf{1}^\perp$; it is $\mathbb{T}$-stable by (H1), (H2). Fix a prime ℓ, a finite extension of $\mathbb{Q}_\ell$ with ring of integers $\mathcal{O}$, uniformiser λ and residue field k; write $\mathbb{T}_\mathcal{O} = \mathbb{T} \otimes \mathcal{O}$ and let $\mathfrak{m} = (\lambda,\ T - N_T : T \in \mathbb{T})$ be the *Eisenstein maximal ideal*. Since $\mathbb{T}_\mathcal{O}$ is a finite commutative $\mathcal{O}$-algebra, $\mathcal{O}^X = \bigoplus_\mathfrak{n} (\mathcal{O}^X)_\mathfrak{n}$ over its maximal ideals; the summand $M_\mathfrak{m}$ is the *Eisenstein component*, the part on which $\mathbb{T}$ acts through eigen-systems congruent to $\chi_{\mathrm{Eis}} : T \mapsto N_T$ modulo λ. Note $\mathbf{1} \in M_\mathfrak{m}$. Put $C_\mathfrak{m} = (C \otimes \mathcal{O}) \cap M_\mathfrak{m}$.

Theorem 2.1 (Martin–Wakatsuki [22], Theorem 2.1 and Lemma 3.4; form-level statement)**.** *For a prime ℓ the following are equivalent:*

(i) $\ell \mid E$;
(ii) *there is $f \in C$ with $f \equiv \mathbf{1} \pmod{\ell}$ coordinatewise;*
(iii) *there is $f \in C_\mathfrak{m}$ with $f \equiv c\mathbf{1} \pmod{\lambda}$ for some $c \in \mathcal{O}^\times$.*

If they hold, there is a Hecke eigenform $f \in C \otimes \overline{\mathbb{Q}}_\ell$ with $\chi_f(T) \equiv N_T \pmod{\lambda}$ for every $T \in \mathbb{T}$.

Proof. (i)⇔(ii). Let $\varphi : \mathbb{Z}^X \to \mathbb{Z}$, $\varphi(f) = \sum_i (e_i/g) f_i$. Since the e_i/g are coprime, φ is surjective; $C = \ker\varphi$ and $\varphi(\mathbf{1}) = E$. If $\ell \mid E$ choose u with $\varphi(u) = -E/\ell$ and put $f = \mathbf{1} + \ell u \in C$. Conversely if $f = \mathbf{1} + \ell u \in C$ then $0 = E + \ell\varphi(u)$.

(ii)⇒(iii). Let $f_0 = \mathbf{1} + \ell u \in C$ and let $e_\mathfrak{m}$ be the idempotent of the Eisenstein component. It is a polynomial in the Hecke operators, so it preserves $C \otimes \mathcal{O}$, and $e_\mathfrak{m}\mathbf{1} = \mathbf{1}$ since $\mathbf{1}$ lies in $M_\mathfrak{m}$ and in no other component. Hence $e_\mathfrak{m} f_0 = \mathbf{1} + \ell e_\mathfrak{m} u \in C_\mathfrak{m}$ and $e_\mathfrak{m} f_0 \equiv \mathbf{1}$.

(iii)⇒(i). Extend φ $\mathcal{O}$-linearly. Then $\varphi(f) = 0$ and $f - c\mathbf{1} \in \lambda\mathcal{O}^X$ give $cE \in \lambda\mathcal{O}$, so $E \in \lambda\mathcal{O} \cap \mathbb{Z} = \ell\mathbb{Z}$.

For the last assertion, the image $\bar{C}$ of $C \otimes \mathcal{O}$ in k^X is $\mathbb{T}$-stable and by (ii) contains $\bar{\mathbf{1}}$, an eigenvector with eigen-system $\bar{\chi}_{\mathrm{Eis}}$. The Deligne–Serre lemma [7, Lemme 6.11] applied to the free $\mathcal{O}$-module $C \otimes \mathcal{O}$ with the commuting operators $\mathbb{T}$ gives an eigenvector in $C \otimes K'$, K' a finite extension of the fraction field, whose eigen-system lifts $\bar{\chi}_{\mathrm{Eis}}$. □

Proposition 2.2. *Write $M = N/D$ in lowest terms. Then $D \mid L$, $Eg = (L/D)\,N$, and*

$$\{\ell : \ell \mid N\} \ \subseteq\ \{\ell : \ell \mid E\} \ \subseteq\ \{\ell : \ell \mid N\} \cup \{\ell : \ell \mid L/D\}.$$

Proof. $D \mid L$ since M is a sum of reciprocals of divisors of L. The identity is the definition of E. If $\ell \mid N$ then $\ell \mid Eg$, and $\ell \nmid g$ because the class attaining the full ℓ-power of L has $\ell \nmid e_i$; so $\ell \mid E$. The right-hand inclusion is immediate from the identity. □

Thus the primes of the mass numerator are always Eisenstein primes, and the only other possible Eisenstein primes are those at which some stabiliser has more ℓ-torsion than the mass denominator records. The second case occurs: at G_2 level 7 the mass is $817/63$ and $E = 2 \cdot 19 \cdot 43$, and the mod-2 congruence exists (§4). It is a feature of the integral statement, not a defect.

2.2. **Three notions of congruence.** It is essential to distinguish, in increasing strength:

(A) a *single-operator coincidence*: $\lambda_T(f) \equiv N_T \pmod{\ell}$ for one Hecke operator T;
(B) a *Hecke congruence*: $\lambda_T(f) \equiv N_T \pmod{\lambda}$ for every $T \in \mathbb{T}$ (the notion of [22]);
(C) a *form-level congruence*: the constant vector lies in the reduction modulo λ of the saturated eigenlattice of the Galois orbit of f, equivalently (Theorem 2.1(iii)) $\bar{\mathbf{1}} \in \overline{C_{\mathfrak{m}}}$.

Clearly (C)⇒(B)⇒(A). Coincidences of type (A) are common at primes $\ell \nmid E$ and are not Eisenstein congruences; every test in §§4–6 is of type (C), and every prime reported as a carrier passes it.

Remark 2.3. (a) Proposition 2.2 is what makes the theorem predictive through the mass formula (1).

(b) Read on the counting side, the equivalence (i)⇔(ii) is a statement about integrality. For a Niemeier lattice L one has $\theta_L = E_{12} + c_L\Delta$ with $c_L = 24h(L) - 65520/691$, so the number of vectors of norm $2n$ is $(65520/691)\sigma_{11}(n) + c_L\tau(n)$; this is an integer because the vectors exist, and its integrality is *equivalent* to $\tau(n) \equiv \sigma_{11}(n) \pmod{691}$. In general the constant function is the ensemble average, whose "counts" (the Eisenstein series) carry a denominator; an actual class carries integer counts; the difference is cuspidal and must repair the denominator, which is the congruence. The primes of the mass are the primes at which the average fails to be integral.

(c) The theorem says nothing about *which* eigenform carries the congruence, nor that the Eisenstein component is a single eigen-system. An eigenvalue coincidence $\lambda_T \equiv N_T \pmod{\ell}$ for a single operator T at a prime $\ell \nmid E$ is common and is not an Eisenstein congruence; the correct test, used throughout §§4–6, is (iii): the constant vector lies in the reduction modulo λ of the saturated eigenspace lattice of a Galois orbit of eigen-systems.

2.3. **Relation to previous mass-congruence theorems.** Martin [21] proved, for definite quaternion algebras, that divisibility of the Eichler mass by p produces a weight-2 Eisenstein congruence, and Martin–Wakatsuki [22] extended this to algebraic modular forms on any reductive group compact at infinity (their Theorem 2.1), with the depth of the congruence bounded below by v_p of the mass [4]; they used it to produce non-endoscopic Eisenstein congruences on unitary groups of prime degree. Bergström–Dummigan [3] treat Eisenstein congruences for maximal parabolics on split groups and their relation to Bloch–Kato. The statement above adds nothing to the existence theorem; what it isolates is the exact integral criterion on the finite class set—the integer $E = LM/g$ governing the constant-vector obstruction, the equivalence with membership of $\bar{\mathbf{1}}$ in the reduced cuspidal Eisenstein component, and the two-inclusion description of Proposition 2.2—in a form that can be checked by machine and used to certify congruences and identify their carriers on the exceptional groups.

3. The mass formula and the predicted primes

Gross [10] showed that groups over $\mathbb{Z}$ with $G(\mathbb{R})$ compact and G split at every prime exist exactly for the types G_2, F_4, E_8 (and suitable classical types), and gave the mass (1). There is no compact $\mathbb{Z}$-form of type E_6 (the compact real form is outer; equivalently $\zeta(1-5) = 0$

kills the mass) or E_7 (the compact form is inner but not pure inner; the two $\mathbb{Z}$-groups of type E_7 classified in [9] are both non-compact at infinity). Combining (1) with Theorem 2.1 and Remark 2.3(a):

Corollary 3.1. *Every prime dividing the numerator of $M(K_S)$ is an Eisenstein congruence prime of level K_S in the sense of Theorem 2.1(iii). In particular:*

- *for the compact G_2 at Iwahori level p: the primes of $(p^2-1)(p^6-1)/(p-1)^2$ not absorbed by $12096 = |G_2(\mathbb{F}_2)|$;*
- *for the compact F_4 at level one:* 691*; at the parahoric level Q_3 (§5):* 41, 73, 691*;*
- *for the compact E_8 at level one, whose class number is at least* 13935 *and whose class set has never been enumerated:* $691,\ 43867,\ 283\cdot 617,\ 103\cdot 2294797,\ 1721\cdot 1001259881$*;*
- *for the compact* SO(24) *(the Niemeier genus):* $131, 283, 593, 617, 691, 3617, 43867$.

4. The compact G_2 at Iwahori level p

4.1. **Construction.** Let $\mathcal{O}$ be the Coxeter order of integral octonions and $\Gamma = \mathrm{Aut}(\mathcal{O}) = G_2(\mathbb{Z})$, of order 12096. Following Lansky–Pollack [15], the class set at Iwahori level p is the set of Γ-orbits on the flag variety of G_2 over $\mathbb{F}_p$: flags (line $\subset$ plane) of null trace-zero elements of $\mathcal{O}/p\mathcal{O}$ with $uv = vu = 0$. There are $(p^6-1)/(p-1)$ null lines and as many planes, each plane containing $p+1$ lines, so $(p+1)(p^6-1)/(p-1)$ flags, and the mass is $\frac{(p+1)(p^6-1)}{(p-1)\cdot 12096}$. The Hecke operator T_q for $q \neq p$ counts, for a representative flag, the images of the $q(q^6-1)/(q-1)$ q-neighbour maps of $\mathcal{O}$ (Gross's $T(\omega_1^\vee)(q)$; here 126, 1092 and 19530 for $q = 2, 3, 5$) reduced modulo p, landing in each class.

p	flags	h	$\sum_x 1/\lvert\Gamma_x\rvert$	E (= Eisenstein primes)
7	156,864	29	817/63	$2\cdot 19\cdot 43$
11	2,125,872	187	703/4	$19\cdot 37$
13	5,631,276	523	67039/144	$7\cdot 61\cdot 157$
17	27,154,764	2280	35919/16	$3^2\cdot 13\cdot 307$

Table 1. Compact G_2 at Iwahori level p. In each case $\sum_x 1/|\Gamma_x|$ equals Gross's mass exactly; every orbit size divides 12096; T_2, T_3 (and T_5 for $p \leq 13$) have the required row sums and mass-weighted symmetry. The class number 29 agrees with [15]; the others are new.

4.2. **Results.** The Eisenstein numbers follow the corollary in detail, including the cancellations: at $p = 11$ the single factor 7 of 11^6-1 cancels against 12096; at $p = 13$ a second 7 enters through $p+1 = 14$ and survives; at $p = 17$ two 3's enter through $p+1 = 18$.

4.3. **Carriers.** For each p and each prime $\ell \mid E$ we determined the Galois orbit of eigen-systems carrying the congruence, by the test of Remark 2.3(c) applied to the factors of the characteristic polynomial of T_2 (computed exactly with FLINT). Three conjectures suggested by the first two levels—that the carrier's Galois orbit has trace 81; that ℓ ramifies in the carrier's coefficient field; that the prime entering through $p+1$ is carried by the large orbit—are each refuted at the next level. What governs the carriers is the following.

p	ℓ	carrier of the form-level congruence
7	43	rational, $(\lambda_2,\lambda_3)=(-3,60)$, multiplicity 3 (the T_2-eigenvalue -3 has multiplicity 9, splitting as $(-3,60)^3\oplus(-3,-4)^6$)
7	19	quadratic orbit $x^2-81x+1512$, multiplicity 2
7	2	genuine (Remark 2.3(a)); joint Eisenstein eigenspace mod 2 of dimension 9
11	37	rational, $(\lambda_2,\lambda_3)=(15,56)$, multiplicity 3
11	19	cubic orbit $x^3-81x^2+1377x+7263$, multiplicity 2
13	157	quadratic orbit $x^2-39x+342$, multiplicity 3
13	61	cubic orbit $x^3-102x^2+2565x-2592$, multiplicity 2
13	7	neither the rational form $\lambda_2=-14$ nor the quartic orbit (its T_3-eigenvalues are not $\equiv 1092$); by elimination the orbit of degree 385
17	13	rational, $(\lambda_2,\lambda_3)=(9,156)$, multiplicity 3
17	3	rational, $(\lambda_2,\lambda_3)=(-18,-60)$, multiplicity 1
17	307	quartic orbit $x^4-117x^3+4257x^2-45063x-36774$ (Galois group S_4, 307 unramified), multiplicity 2; its two copies are the whole cuspidal Eisenstein component at 307

TABLE 2. Which eigenforms carry the Eisenstein congruences on G_2.

4.4. **The carriers are Arthur-type lifts.** Let $\mathrm{SO}(4)=(\mathrm{SL}_2^{\mathrm{long}}\times\mathrm{SL}_2^{\mathrm{short}})/\mu_2\subset G_2$ be the stabiliser of a quaternion subalgebra of the octonions, so that the 7-dimensional representation restricts as $(2,2)\oplus(1,3)$. For a newform $f\in S_6(\Gamma_0(p))$ consider the Arthur-type parameter $\psi_f=\varphi_f\boxtimes\nu$ with φ_f on the long-root SL_2 and the Arthur SL_2 on the short root. Its Satake parameter at $q\neq p$ has $\chi_7=(q+1+q^{-1})+(q^{1/2}+q^{-1/2})\,a_q(f)q^{-5/2}$, and since the eigenvalue of $T(\omega_1^\vee)(q)$ on a trivial-weight form is $q^3\chi_7-1$ (the Eisenstein eigenvalue $q^6+\cdots+q$ and the lift of 3.6.a.a at level 3, $(T_2,T_5,T_7)=(9,810,2472)$, both verify this normalisation),

$$T_q(\psi_f)=q^4+q^3+q^2-1+(q+1)\,a_q(f),\qquad T_q(\mathrm{Eis})-T_q(\psi_f)=(q+1)\big(1+q^5-a_q(f)\big).$$

Theorem 4.1. *At G_2 Iwahori level $p\in\{7,11,13,17\}$ every newform $f\in S_6(\Gamma_0(p))$ occurs as ψ_f in the class set, with multiplicity 3 or 2 according to its Atkin–Lehner sign at p (7, 9, 12, 14 eigen-slots in all); these lifts are non-tempered and are exactly the eigen-systems violating the tempered bound $|T_q+1|\le 7q^3$ at $q=3$ ($p=7,11,17$) or $q=2$ ($p=13$). The G_2 Eisenstein congruence of ψ_f modulo ℓ is the classical congruence $f\equiv E_6\pmod{\ell}$, and every cyclotomic carrier of Table 2 is such a lift: the prime dividing p^2-p+1 (43, 37, 157, 13) is carried by the multiplicity-3 lift and the prime dividing p^2+p+1 (19, 19, 61, 307) by the multiplicity-2 lift of the newform congruent to E_6 modulo that prime (verified in the coefficient fields at $q\le 11$). The primes from $p\mp 1$ are carried by tempered forms: 3 at $p=17$ by $(-18,-60)$, which is no lift ($a_2=-15$ would exceed the Deligne bound), and 2 at $p=7$ by the genuine component of dimension 9.*

Proof. Exact eigenvalue matching of the predicted (T_2,T_3) against the joint spectra of the Hecke matrices, newform coefficients from PARI; the carrier orbits of Table 2 are the Galois orbits of the lifts (their traces $81,81,102,117$ are $\sum(27+3a_2)$ over the orbit, which is what the refuted "trace 81" conjecture had noticed), and the classical congruences were checked by norms in the coefficient fields. The dictionary itself is proved exactly at level 3 in Part IV, where the same lifts in weight $2\omega_1$ are matched to the weight-10 newforms of level 3. □

Thus the cyclotomic part of the law is classical Eisenstein theory transported through the non-tempered lifts, exactly as 73 and 41 were on F_4 (Section 5); what the lifts do not explain are the Bernoulli-type primes (7 for G_2, 691 for F_4) and the $p\mp1$ primes, which fall on tempered forms. The tempered remainders have dimensions 21, 177, 510, 2265.

Proposition 4.2 (the $p\mp1$ primes are stable)**.** *A tempered endoscopic parameter through* SO(4) *in trivial weight requires a weight-*6 *form on the long-root* SL_2 *and a weight-*2 *form on the short-root* SL_2*, both of level dividing* p*. Since* $S_2(\Gamma_0(7)) = S_2(\Gamma_0(13)) = 0$*, the carriers of* 2 *at level* 7 *and of* 7 *at level* 13 *are stable. At level* 17 *the pairs* (f, g) *with* g *the newform of weight* 2 *and* $f \in S_6(\Gamma_0(17))$ *have predicted eigenvalues* $T_2 = -a_2(f) - 5$*,* $T_3 = -28$*; the four pairs from the quartic Galois orbit of* f *are present in the class set with multiplicity* 1*, the two pairs from the rational forms are absent, and the rule is* $w_{17}(f) = w_{17}(g)$ *(the local sign condition of the multiplicity formula). The carrier* $(-18, -60)$ *of* 3 *is none of these (*$a_2(f) = 13$ *would exceed the Deligne bound), hence stable. So the primes dividing* $p\mp1$ *fall on forms with no classical origin at all four levels.*

4.5. **A blind test at level** 19**.** Theorem 4.1 was written on 4 October 2026 from the levels 7–17. On 8 October, before any computation at 19, we predicted from $S_6(\Gamma_0(19))$ (four newforms: rational with $a_2 = -2$ and $w = +1$, rational with $a_2 = -6$ and $w = -1$, a quadratic pair with $w = +1$, a quartic orbit with $w = -1$) that the Iwahori class set at 19 contains 19 lift slots with the eigenvalues $T_2 = 27 + 3a_2$ of multiplicities $3, 2, 3, 3, 2, 2, 2, 2$; that the Eisenstein primes are $5, 7, 127$ (mass numerator $5^2 \cdot 7^2 \cdot 127$, with $127 \mid \Phi_3(19)$, $7 \mid \Phi_6(19) = 7^3$ and $5 \mid 20$); that 127 is carried by the quartic lifts (multiplicity 2), 7 by the two $w = +1$ lifts (multiplicity 3), and, for the first time among the $p\mp1$ primes, 5 by a lift, since the form with $a_2 = -2$ is congruent to E_6 modulo 5 as well as 7. The class set was then computed with a rebuilt implementation (validated on the class numbers $29, 187, 523$, the exact masses, and the level-7 spectrum to the last multiplicity): $h(19) = 4437$, mass $155575/36$ exactly, T_2 with row sums 126 and exact mass symmetry. Seven of the eight predicted eigenvalues appear with exactly the predicted multiplicities ($54.085, 49.131, 32.549, -0.764$ twice each; $36.456, -3.456, 21$ three times each); the eighth, $T_2 = 9$, appears three times against a predicted two, a coincidence with a tempered form to be separated by T_5 (predicted lift pairs $(T_2, T_5) = (21, 630)$, $(9, 1098)$, $(-3.456, 175.438)$, $(36.456, 574.562)$, $(32.549, 835.663)$, $(54.085, 330.978)$, $(49.131, 1425.087)$, $(-0.764, 1044.272)$). The structural content of the theorem thus holds blind at a fifth level.

4.6. **Multiplicities and local types.** The multiplicity of a joint (T_2, T_3) eigen-system on the class set at Iwahori level p is the dimension of the space of Iwahori-fixed vectors of the local representation at p (for eigen-systems of global multiplicity one), so it records the local type of the form at the level prime. Among the rational eigen-systems we find only the multiplicities $1, 2, 3, 6$: at $p = 7$: $(-14, -48)^3, (-3, 60)^3, (-3, -4)^6$; at $p = 11$: $(-9, 12), (5, 36)$ with multiplicity 1, $(-6, -7), (14, 3)$ with 2, $(15, 56), (-6, 13 \pm \sqrt{37})$ with 3; at $p = 13$: $(-14, -63)$; at $p = 17$: $(-18, -60), (6, 36), (6, -28)$ with 1 and $(4, -27), (9, 156), (30, 44)$ with 3. The carriers of the cyclotomic Eisenstein primes 43, 37 and 13 all have multiplicity 3 and 3 (at $p = 17$) has multiplicity 1; Theorem 4.1 explains the multiplicities as the Iwahori-fixed dimensions of the local components of the Arthur-type lifts, decided by the Atkin–Lehner sign of the classical form.

4.7. **Congruence numbers.** The index $[\mathbb{Z}^X : \bigoplus_\chi (\text{saturated eigenlattice of } \chi)]$ is the discriminant of the Hecke module; its prime factorisation lists every congruence prime with a multiplicity. For F_4 at level Q_3 it is

$$2^{16} \cdot 3^{11} \cdot 5^6 \cdot 7 \cdot 11^3 \cdot 13 \cdot 17^2 \cdot 41 \cdot 73 \cdot 691,$$

in which each Eisenstein prime appears exactly once (the Eisenstein number E divides it exactly) and the cuspidal congruences of §5 account for 17^2 (the two-dimensional intersection), 11^3 (two-dimensional plus one), 13 and 7. For the Niemeier genus it is a 240-digit number whose odd part beyond 3 is

$$5^{34}\cdot 7^{20}\cdot 11^{10}\cdot 13^{11}\cdot 17^{3}\cdot 19^{2}\cdot 23^{2}\cdot 41\cdot 131\cdot 283^{2}\cdot 593\cdot 617^{2}\cdot 691^{10}\cdot 3617^{4}\cdot 43867^{3},$$

containing Harder's 41 once and the Bernoulli primes with the multiplicities of their propagation among the lifts.

5. The compact F_4

5.1. **Level one.** The compact F_4 over $\mathbb{Z}$ is $\mathrm{Aut}(J_{\mathbb{Z}})$ for the integral Albert algebra $J_{\mathbb{Z}} = \mathrm{Her}_3(\mathcal{O})$, and its level-one class set has two elements [10, 8]: $J_{\mathbb{Z}}$, with $|\mathrm{Aut}| = 4180377600 = 6|W(E_8)|$, and the Elkies–Gross isotope J_E, with $|\mathrm{Aut}| = 634023936 = 3|{}^3D_4(2)|$; the mass is $691/380414361600$. Their theta series are $E_{12} \pm (\mathrm{const}/691)\Delta$, so the level-one Hecke module is $\mathbb{C}\mathbf{1} \oplus \mathbb{C}\,\Theta(\Delta)$ [26], and the congruence mod 691 is Ramanujan's. We enumerated the 2-neighbours of both lattices for the two maximal parabolics (rank-one trace-zero points of $J/2J$, 69615 of them, in five Aut-orbits of sizes $48600, 14175, 6075, 405, 360$; and 6-dimensional null subspaces for the Heisenberg parabolic) and classified each neighbour by explicit isomorphism. The Brandt matrices at $p = 2$ are

$$T(\omega_4^\vee)(2) = \begin{pmatrix} 1264320 & 7646400 \\ 1159704 & 7751016 \end{pmatrix}, \qquad T(\omega_1^\vee)(2) = \begin{pmatrix} 25830 & 113400 \\ 17199 & 122031 \end{pmatrix},$$

with eigenvalues $\{8910720, 104616\}$ and $\{139230, 8631\}$. Both cuspidal eigenvalues agree with the Satake-parameter prediction from $\tau(2) = -24$ for the lift of Δ via $\iota : \mathrm{Sp}_6 \times \mathrm{SL}_2 \to F_4$ with the principal SL_2 (for $T(\omega_1^\vee)$: $N_2 - (1+2+\cdots+2^5)(1+2^{11}-\tau(2)) = 139230 - 63\cdot 1971 = 15057$ at level 3 below, and 8631 at level one). The 819 rank-one elements of J_E that define its relation to the Leech lattice [8] appear as the orbit of size 360; each has exactly 72 standard neighbours.

5.2. **Level Q_3.** Let Q_3 be the parahoric at 3 attached to $\omega_1^\vee$ (stabiliser of a 6-dimensional null subspace of $J/3J$). The class set has 15 elements: nine on $J_{\mathbb{Z}}$ with stabiliser orders 288, 1296, 1944, 3024, 15552, 23328, 72576, 645120, 1244160, six on J_E with 36, 288, 648, 972, 2304, 9072; and

$$\sum_x \frac{1}{|\Gamma_x|} = \frac{2068163}{52254720} = \frac{691}{380414361600}\cdot 21789040,$$

21789040 being the number of Q_3-cosets. The matrix of $T(\omega_1^\vee)(2)$ (Appendix A) has row sums 139230, is mass-weighted symmetric, and has integer spectrum

$$139230,\quad 8631^{(5)},\quad 15057^{(2)},\quad 3087^{(2)},\quad 2331,\quad 567^{(2)},\quad 207,\quad -585.$$

Here $E = 41\cdot 73\cdot 691$, as (1) predicts ($41 \mid 3^8-1$, $73 \mid 3^{12}-1$, 691 from $\zeta(-11)$, with 5 and 7 cancelling), and the form-level test passes at exactly these primes: 691 on the Δ-lift eigenspace, 73 on the 15057-eigenspace, 41 on the 2331-eigenspace, and at no other prime (the eigenvalue coincidences at $2161, 53, 31, 71, 19, 271, 13, 239$ all fail the test).

5.3. **The two classical lifts and the prime 73.** The theta correspondence for $F_4 \times \mathrm{PGL}_2$ [26] lifts a newform f of weight 12 and level 3 to eigenvalue $N_2 - 63\,(1 + 2^{11} - a_2(f))$. The space $S_{12}(\Gamma_0(3))$ has a unique newform, 3.12.a.a, with $a_2 = 78$, $a_3 = -3^5$; it lifts to 15057, and $1 + p^{11} - a_p(f) \equiv 0 \pmod{73}$ for $p = 2, 5, 7$: the prime 73 on F_4 is the classical Eisenstein congruence of 3.12.a.a, transported, exactly as 691 is Ramanujan's. The remaining eigenvalues are *not* such lifts: $3087, 2331, 567$ would need $a_2 = -112, -124, -152$, outside the Ramanujan bound $2\cdot 2^{11/2}$, and $207, -585$ are not of the required shape at all.

5.4. **Theta-visibility.** The classical theta map of the level-Q_3 classes (rank-one elements counted by trace) is Hecke-equivariant into $M_{12}(\Gamma_0(3))$, which has dimension 5 and T_2-eigenvalues $a_2 \in \{2049, -24, 78\}$ for the Eisenstein space, Δ and 3.12.a.a. By the dictionary $N_2 - 63(2049 - a_2)$ these correspond to the eigenvalues 139230, 8631 and 15057, and no others: the eigenspaces $3087, 2331, 567, 207, -585$ lie in the kernel of the theta map. Consequently the primes 691 and 73 are explained by integrality of rank-one counts in the sense of Remark 2.3(b)—the $\Gamma_0(3)$ Eisenstein combination $(E_{12}(\tau) - 3^{12}E_{12}(3\tau))/(1 - 3^{12})$ has denominator $691 \cdot (3^{12} - 1) = 2^4 \cdot 5 \cdot 7 \cdot 13 \cdot 73 \cdot 691$—while 41 is not: $41 \mid 3^8 - 1$, and $3^8 - 1$ occurs in no weight-12 Eisenstein denominator at level 3. The 41-carrier is invisible to every classical theta series of weight 12; its explanation (Part II [1]) is that it is attached to a newform of weight 8, whose Eisenstein series has $3^8 - 1$ in its denominator.

5.5. **The remaining eigen-systems.** The eigenspaces $3087^{(2)}, 2331, 567^{(2)}, 207, -585$ are the subject of Part II [1], where a second Hecke operator $T(\omega_4^\vee)(2)$ is computed and each eigensystem is matched, at the prime 2, with a parameter on a smaller group: 2331 with the parameter 3.8.a.a[3] on Sp_6 (which explains 41 as the Eisenstein congruence of the weight-8 newform of level 3, transported), 207 with 3.6.a.a $\oplus$ [4] paired with Δ, 3087 and 567 with compact G_2-forms of weight $2\omega_2$ and level 3, and -585 with a compact SO(7)-form of weight $(3, 1, 1)$ on the genus of E_7 whose Frobenius polynomial factors as 3.6.a.a times a stable genus-2 parameter. Every prime of the congruence number of §4 is thereby explained either as a transported classical Eisenstein congruence or as a level-raising congruence at 3.

6. The Niemeier genus

Chenevier and Lannes [6] publish the Kneser neighbour counts $N(p, L_i, L_j)$ between the 24 Niemeier lattices for $p \le 31$. Assembling them into Hecke matrices T_p, the weights are determined by (H2) up to a scalar fixed by the Smith–Minkowski–Siegel mass; they come out integral, with $|\operatorname{Aut}(\text{Leech})| = |\mathrm{Co}_0|$, and $E = 131 \cdot 283 \cdot 593 \cdot 617 \cdot 691^2 \cdot 3617 \cdot 43867$. Of the 23 cuspidal T_2-eigenforms (all rational, [24]), exactly seven are congruent to the constant function: 691 on the Ikeda lift of Δ ($\lambda_2 = 4192830$) and on two further forms; $131 \cdot 593$ (numerator of B_{22}) on $\lambda_2 = 2098332$; $283 \cdot 617$ (B_{20}) on 533160; 43867 (B_{18}) on 99792; 3617 (B_{16}) on 89640. This is the set predicted by Corollary 3.1 with nothing missing and nothing extra; Harder's prime 41, which is a Siegel–elliptic congruence on this genus, is correctly absent. The 691-eigenform is the Coxeter-number function $L \mapsto 691\, h(L) - 2730$, whose value at the Leech lattice is the denominator of B_{12}.

6.1. **The umbral groups are the weights.** The weights of the Hecke inner product factor in a way that connects the theorem to umbral moonshine [5]. For each of the 23 Niemeier lattices L^X with root system X, Cheng, Duncan and Harvey define the *umbral group* $G^X = \operatorname{Aut}(L^X)/W(X)$, the quotient by the Weyl group, and attach to X a vector-valued mock modular form whose index is the Coxeter number $h(X)$ (their lambency). Dividing the automorphism orders recovered from (H2) and the mass by the Weyl group orders $\prod_i |W(X_i)|$ reproduces their table in all 23 cases: M_{24} for A_1^{24}, $2.M_{12}$ for A_2^{12}, $2.\mathrm{AGL}_3(2)$ for A_3^8, $\mathrm{GL}_2(5)/2$ for A_4^6, $3.\mathrm{Sym}_6$ for D_4^6, $\mathrm{GL}_2(3)$ for E_6^4 and $A_5^4D_4$, Sym_3 for E_8^3, and so on down to the trivial group for $D_{16}E_8$ and D_{24} (and Co_0 for the Leech lattice, where W is trivial). Hence

$$\langle f, g \rangle = \sum_X \frac{f(X)\, g(X)}{|W(X)| \cdot |G^X|},$$

and the Eisenstein number E, hence the seven Eisenstein primes, is computed from the Weyl groups and the umbral groups. In the same variable two of the Eisenstein eigenforms become

explicit polynomials: with $h = h(X)$,

$$v_{691}(X) = 2730 - 691\,h, \qquad v_{131\cdot593}(X) = 77683\,(7h^2 - 60h) + 9418500, \quad 77683 = 131 \cdot 593,$$

each congruence visible coefficient by coefficient, and each constant term—the value at the Leech lattice, $h = 0$—carrying the denominator of the corresponding Bernoulli number (2730 for B_{12}; $9418500 = 138 \cdot 68250$ for B_{22}). The 3617, 43867 and $283 \cdot 617$ eigenforms are not polynomials in h, nor in h together with the number and types of the components; in particular the 3617-eigenform does not vanish at E_8^3, whose root system is a 7-design, so it is not the degree-4 harmonic theta vector. We do not know what they are as functions of the root system.

6.2. **The character map and the vertex operator algebras.** The theta map $\Theta\colon f \mapsto \sum_X f(X)\theta_X/|\operatorname{Aut} X|$ from functions on the genus to M_{12} is Hecke-equivariant and sends the constant function to E_{12} and v_{691} to Δ. Dividing by Δ identifies M_{12} with $\operatorname{span}\{J, 1\}$, $J = j-744$, which by Zhu's theorem is exactly the space of characters of strongly rational holomorphic vertex operator algebras of central charge 24: every such V has character $J + \dim V_1$, where V_1 is its weight-one Lie algebra. Under this map the 691-eigenline becomes the constant direction, i.e. the functional $\dim V_1$, while the Eisenstein line becomes $E_{12}/\Delta = J + 82104/691$, the character of no vertex operator algebra: Ramanujan's congruence, read on the VOA side, is the statement that E_{12} fails to be a character and fails by 691. For a Niemeier lattice, $\dim(V_X)_1 = 24(h(X)+1)$, so the Eisenstein eigenform extends to all 71 vertex operator algebras of Schellekens' list as

$$v(V) = 3421 - \tfrac{691}{24}\dim V_1,$$

and at the Moonshine module, $V_1^\natural = 0$, it takes the value $3421 = 2730 + 691$: one unit of 691 beyond the Leech lattice vertex operator algebra, $V^\natural \equiv V_\Lambda \pmod{691}$.

Höhn [13], Möller–Scheithauer [16] and Höhn–Möller [14] show that the 70 vertex operator algebras with $V_1 \neq 0$ fall into eleven families, each a simple-current extension built on a lattice from a fixed classical positive-definite genus (the Niemeier genus $II_{24,0}$; $II_{16,0}(2_{II}^{+10})$ with 17 classes; $II_{12,0}(3^{-8})$ with 6; and so on down to rank 4), with the orbifold constructions realised at the lattice level by passage to index-n sublattices of fixed lattices. Theorem 2.1 therefore applies to the vertex operator algebras family by family, with the Eisenstein primes of each family being the primes of the Siegel mass of its genus; the Niemeier family is the case treated above. The Moonshine module is the rank-zero case: its orbit lattice is the zero lattice, its family is the 38 rank-zero generalised deep holes of V_Λ, and its "genus" is a single point of mass 1, which has no prime numerator. No Eisenstein congruence reaches $V^\natural$ through this structure.

Remark 6.1 (Averaged holography)**.** The mass-weighted average of the partition functions θ_X/η^{24} of the Niemeier lattice vertex operator algebras is E_{12}/Δ (Siegel–Weil); in the "averaged holographic duality" of Maloney–Witten and Afkhami-Jeddi–Cotler–Hartman–Maloney–Tajdini [17, 18], such an Eisenstein series is the partition function of a simple three-dimensional gravity theory summed over its saddles, and in Witten's proposal [20] the Moonshine module, with no weight-one currents, is pure gravity at $c = 24$. In our data the mass-weighted average of $\dim V_1$ over the 24 Niemeier theories is exactly 82104/691: the ensemble-averaged chiral $c = 24$ lattice theory has 118.82 currents, is not a theory, and the denominator of its current count is Ramanujan's prime. In this language the constant function on a genus is the averaged theory, the cusp forms are the deviations of individual theories from it, and Theorem 2.1 says that an individual theory is congruent to the ensemble average modulo ℓ exactly when ℓ divides the numerator of the mass. We record this as a reading, not a result: congruences of partition functions modulo primes have, at present, no physical interpretation.

We note what this bridge is not. Monster moonshine is the moonshine of $V^\natural$, the $\mathbb{Z}/2$-orbifold of the Leech lattice vertex algebra, with character $j - 744$; the coefficients of j carry congruences

at the supersingular primes $2, \dots, 13$ and at no Bernoulli prime, the character degrees of the Monster divide its order and so involve no Bernoulli prime, and the coefficients of the umbral mock modular forms are dimensions of representations of the umbral groups. Every Bernoulli-prime phenomenon in this paper lives on the lattice and Hecke side of the Leech lattice; none crosses to the group side.

7. The compact E_8, and the absent E_6 and E_7

The ladder of compact groups over $\mathbb{Z}$ built from the octonions has fewer rungs than the exceptional series suggests. There is no compact $\mathbb{Z}$-form of type E_6: the compact real form is an outer form, so a $\mathbb{Q}$-group split at every finite prime and compact at infinity would need a quadratic twist trivial at every finite place and nontrivial at infinity, which does not exist; the mass formula records the same fact, since E_6 has the odd Weyl degrees 5 and 9 and $\zeta(1-5) = 0$. There is no compact $\mathbb{Z}$-form of type E_7 either: the compact form is inner but not pure inner, and the two $\mathbb{Z}$-groups of type E_7 classified in [9] (the automorphism groups of the Freudenthal triple systems over $\mathrm{Her}_3(C)$ for C the split octonions and the Coxeter order) are both non-compact at infinity; the octonionic one is the Hermitian group $E_{7,3}$ studied by Pollack. So after F_4 the next compact rung is E_8.

Gross [10] and Gross–Nebe [11] describe its integral structures: the unique anisotropic E_8 over $\mathbb{Q}$ is split at every prime, and its level-one class set is the set of isomorphism classes of *Lie orders*—$\mathbb{Z}$-lattices $\mathcal{L}$ in the 248-dimensional Lie algebra $\mathfrak{g}_{\mathbb{Q}}$, closed under the bracket, on which the Killing form divided by $2h^\vee = 60$ is even, unimodular and negative definite—that are hyperspecial at every prime. One class is explicit: the Thompson–Kostrikin orthogonal decomposition of $\mathfrak{e}_8$ into 31 mutually orthogonal Cartan subalgebras gives a Lie order $\mathcal{L}_{\mathrm{OD}} \cong_\perp E_8^{31}$; the stabiliser of the decomposition in $E_8(\mathbb{C})$ is Alekseevskii's group $2^{5+10} \cdot \mathrm{SL}_5(2)$, of order 327,659,028,480, and it contains the Dempwolff group $2^5 \cdot \mathrm{GL}_5(2)$, of order 319,979,520. The mass is

$$M(E_8) = 2^{-8} \prod_{d \in \{2,8,12,14,18,20,24,30\}} |\zeta(1-d)| = \frac{2155741910416889170788798426985697}{154705492508859411569049600000} \approx 13934.9,$$

so the class number is at least 13935 [10, Prop. 5.3]; the Thompson–Kostrikin class contributes at most $3.2 \cdot 10^{-9}$ to the mass, and the class set is dominated by classes with tiny automorphism groups, as the Niemeier genus is dominated by its Leech end but incomparably more so. The class number is unknown.

Theorem 2.1 (that is, Martin–Wakatsuki's theorem) nevertheless applies. Its hypotheses are the finiteness of the class set and the mass formula, both proved by Gross, and the two matrix properties (H1), (H2), which hold for every Hecke operator on every class set. Hence:

Theorem 7.1. *On the space of level-one algebraic modular forms of trivial weight on the compact E_8 over $\mathbb{Z}$, for each of the primes*

$$691, \quad 43867, \quad 283, \quad 617, \quad 103, \quad 2294797, \quad 1721, \quad 1001259881$$

—the primes of the numerators of B_{12}, B_{18}, B_{20}, B_{24}, B_{30}—there is an integer-valued cuspidal function congruent to the constant function modulo ℓ, an element of the cuspidal Eisenstein component congruent to a unit multiple of the constant, and a cuspidal Hecke eigenform whose eigen-system is congruent to the Eisenstein one modulo a prime above ℓ. No other prime has this property except possibly primes dividing $\operatorname{lcm} |\Gamma_x|$.

The prime 3617 (numerator of B_{16}) is absent because 16 is not a Weyl degree of E_8; it appears on the Niemeier genus (§6) because 16 is a degree of D_{12}.

We have not attempted to enumerate this class set. The instrument that handles G_2 (an 8-dimensional lattice, 63 rank-one points modulo 2) and F_4 (27 dimensions, 69615 points, 15 classes at level Q_3) finds isomorphisms between lattices by backtracking on rank-one elements, which works because the automorphism groups involved are large. For tens of thousands of 248-dimensional Lie orders with automorphism groups of order 1 or 2 there is no such handle, and isomorphism testing at that scale is a research problem in its own right. Theorem 7.1 is therefore a statement about a Hecke module whose dimension is unknown, obtained from four values of the zeta function: the congruence-prime spectrum of the module is known although the module itself has never been constructed. We regard this, and the identification of the unknown carriers on F_4, as the principal open experiments the paper leaves.

The subgroup-theoretic relation between the Elkies–Gross lattice, the Thompson group and $E_8(\mathbb{F}_3)$ is recorded in Appendix D.

8. Concluding remarks

Theorem 2.1, due in substance to Martin and Wakatsuki [22], is Hida's principle—congruence primes are primes of the adjoint L-value—for the trivial representation, where the adjoint L-value is the mass; its content is that the principle holds at the level of functions on the class set, with a proof short enough to formalise. They applied it to unitary groups and to SO(5); our contribution is its application, with identified carriers, to the exceptional groups and the Niemeier genus. The data confirm it on three groups and eight levels and, on the Niemeier genus, against computations we did not make. The questions the data leave open are of two kinds: the automorphic identification of the F_4 and G_2 eigenforms, pursued in Part II [1]; and the Bernoulli-type and $p \mp 1$ Eisenstein primes, which fall on tempered forms and are not explained by the lifts of Theorem 4.1.

Appendix A. The Hecke matrix at F_4 level Q_3

Classes 1–9 are on $J_{\mathbb{Z}}$ and 10–15 on J_E, in the order of the stabiliser lists above. Entry (i,j) counts the $\omega_1^\vee$-neighbours at 2 of class i lying in class j.

$$\begin{pmatrix}
18270 & 3528 & 2268 & 1344 & 189 & 189 & 42 & 0 & 0 & 92232 & 10920 & 5040 & 3528 & 1344 & 336 \\
15876 & 7011 & 0 & 2808 & 0 & 0 & 135 & 0 & 0 & 93744 & 9504 & 5400 & 3024 & 1296 & 432 \\
15309 & 0 & 9279 & 0 & 567 & 675 & 0 & 0 & 0 & 91692 & 11340 & 3888 & 4212 & 2268 & 0 \\
14112 & 6552 & 0 & 4851 & 0 & 0 & 252 & 63 & 0 & 84672 & 19656 & 7560 & 0 & 756 & 756 \\
10206 & 0 & 4536 & 0 & 10521 & 378 & 0 & 0 & 189 & 81648 & 18144 & 0 & 9072 & 4536 & 0 \\
15309 & 0 & 8100 & 0 & 567 & 1854 & 0 & 0 & 0 & 83592 & 13608 & 7776 & 8424 & 0 & 0 \\
10584 & 7560 & 0 & 6048 & 0 & 0 & 1449 & 189 & 0 & 84672 & 12096 & 3024 & 0 & 4536 & 9072 \\
0 & 0 & 0 & 13440 & 0 & 0 & 1680 & 8064 & 2646 & 0 & 94080 & 0 & 0 & 5880 & 13440 \\
0 & 0 & 0 & 0 & 15120 & 0 & 0 & 5103 & 5607 & 0 & 90720 & 0 & 0 & 22680 & 0 \\
11529 & 2604 & 1698 & 1008 & 189 & 129 & 42 & 0 & 0 & 99945 & 11214 & 5568 & 3582 & 1386 & 336 \\
10920 & 2112 & 1680 & 1872 & 336 & 168 & 48 & 42 & 21 & 89712 & 21015 & 5280 & 3024 & 2376 & 624 \\
11340 & 2700 & 1296 & 1620 & 0 & 216 & 27 & 0 & 0 & 100224 & 11880 & 8415 & 864 & 108 & 540 \\
11907 & 2268 & 2106 & 0 & 567 & 351 & 0 & 0 & 0 & 96714 & 10206 & 1296 & 10413 & 3402 & 0 \\
10752 & 2304 & 2688 & 576 & 672 & 0 & 144 & 21 & 42 & 88704 & 19008 & 384 & 8064 & 5679 & 192 \\
10584 & 3024 & 0 & 2268 & 0 & 0 & 1134 & 189 & 0 & 84672 & 19656 & 7560 & 0 & 756 & 9387
\end{pmatrix}$$

Appendix B. The Lean certificate

The file `Eisenstein.lean` (Lean 4 v4.34.0, Mathlib v4.34.0) proves, with no `sorry` and using only the axioms `propext`, `Classical.choice`, `Quot.sound`:

- `mass_criterion`: for $e, c : X \to \mathbb{Z}$ with $\sum e_i c_i = 1$, $(\exists f,\ \sum e_i f_i = 0 \wedge \forall i,\ \ell \mid f_i - 1) \iff \ell \mid \sum e_i$;
- `cuspidal_stable`: a matrix with constant row sums and $e_i T_{ij} = e_j T_{ji}$ preserves $\{f : \sum e_i f_i = 0\}$;
- `eisenstein_component_to_mass`: for ℓ prime, $\ell \nmid c$, f cuspidal with $\ell \mid f_i - c$ for all i, $\ell \mid \sum e_i$.

The two remaining steps of the proof of Theorem 2.1 are formalised in the file `EisensteinComponent.lean` (same toolchain; compiled by GitHub Actions, public log at https://github.com/chrisyounge-create/Cayley-Agent-Repository), again with no `sorry` and the same three axioms:

- `eisenstein_idempotent_step`: given a complete family of orthogonal idempotents e_j of the commutative algebra $\mathbb{T}_{\mathcal{O}}$ acting on $\mathcal{O}^X$ (its existence for a finite algebra over a complete discrete valuation ring is the standard decomposition into local factors and is taken as a hypothesis), an eigenvector $\mathbf{1}$ with character $\chi_{\mathrm{Eis}} : \mathbb{T}_{\mathcal{O}} \to \mathcal{O}$, a $\mathbb{T}_{\mathcal{O}}$-stable submodule C, and $f_0 \in C$ with $f_0 \equiv \mathbf{1} \pmod{\ell}$: some e_j fixes $\mathbf{1}$, and $e_j f_0$ lies in C, in the component $e_j\mathcal{O}^X$, and is $\equiv \mathbf{1} \pmod{\ell}$;
- `deligne_serre_prime` and `deligne_serre_lift`: for a torsion-free algebra A over a domain $\mathcal{O}$ and a prime $\mathfrak{n}$ of A, there is a prime $P \subseteq \mathfrak{n}$ with $P \cap \mathcal{O} = 0$, so that $A \to A/P$ is a ring homomorphism to a domain, injective on $\mathcal{O}$, reducing modulo $\mathfrak{n}/P$ to the given character.

Applied to $A = \mathbb{T}_{\mathcal{O}}/\operatorname{Ann}(C \otimes \mathcal{O})$ and $\mathfrak{n}$ the kernel of the Eisenstein character modulo λ, this is the Deligne–Serre lemma in the form used in the proof; the finiteness of $\operatorname{Frac}(A/P)$ over $\operatorname{Frac}(\mathcal{O})$, which follows from the finiteness of A over $\mathcal{O}$, is not restated in the certificate.

Appendix C. Data and code

All code (Python/NumPy, FLINT for exact characteristic polynomials), the class sets, Brandt matrices and characteristic polynomials for G_2 at levels $7, 11, 13, 17$ and F_4 at levels 1 and Q_3, the Niemeier computation and the Lean file are available from the author. The carrier of 307 at G_2 level 17 was found from the characteristic polynomial of the 2280×2280 matrix T_2 (reconstructed from 106 word-size primes and verified against a 107th): the cusp polynomial's Eisenstein factor at 307 is a Hensel-lifted square $(x-\alpha)^2$, $\alpha \in \mathbb{Z}_{307}$, and lattice reduction on α returns the quartic of Table 2, which divides the cusp polynomial exactly twice.

Appendix D. A thread to the Monster through the Thompson group

Section 6 found that no Bernoulli-prime phenomenon reaches the Monster. There is, however, a subgroup-theoretic thread, and it runs through both exceptional class sets at level one. The automorphism group of the Elkies–Gross lattice J_E has order $634{,}023{,}936 = 3\,|{}^3D_4(2)|$ and contains ${}^3D_4(2)$; since the 26-dimensional representation of ${}^3D_4(2)$ is absolutely irreducible and rational, a central element of order 3 would act on $J_E \otimes \mathbb{Q}$ by a rational scalar, hence trivially, so $\operatorname{Aut}(J_E) \cong {}^3D_4(2){:}3 = \operatorname{Aut}({}^3D_4(2))$, not ${}^3D_4(2) \times 3$. This group is a maximal subgroup of the Thompson group Th, the point stabiliser of its minimal permutation representation, of degree $|\mathrm{Th}|/|{}^3D_4(2){:}3| = 143{,}127{,}000 = 2^3 \cdot 3^5 \cdot 5^3 \cdot 19 \cdot 31$ [2]; and $\mathrm{Th} \cong C_{\mathbb{M}}(3C)/\langle 3C \rangle$ is a section of the Monster. On the E_8 side, Th acts on the Thompson–Smith lattice, a $\mathbb{Z}$-form of $\mathfrak{e}_8$ whose bracket it preserves only modulo 3, and the subgroup preserving the bracket over $\mathbb{Z}$ is the Dempwolff group, again a maximal subgroup of Th [19]. Thus *the one explicit non-split point of each exceptional class set—J_E for F_4, a Dempwolff-stable Lie order for E_8—is stabilised by a maximal subgroup of the Thompson group.*

The thread is 3-adic, and 3 is the level prime of §5. The Thompson group embeds in $E_8(\mathbb{F}_3)$ through the $3C$-centraliser of the Monster acting on a vertex algebra over $\mathbb{F}_3$ containing $\mathfrak{e}_8 \otimes \mathbb{F}_3$; reduction modulo 3 embeds $\operatorname{Aut}(J_E)$ in $F_4(\mathbb{F}_3)$ (the kernel of reduction on a finite group is trivial for $p \geq 3$); and the six exotic classes at level Q_3 are precisely the orbits of ${}^3D_4(2){:}3$ on the 21,789,040 points of the $F_4(\mathbb{F}_3)$-flag variety, with stabilisers of orders $36, 288, 648, 972, 2304, 9072$. It is natural to ask whether this is more than an abstract coincidence, i.e. whether $\operatorname{Aut}(J_E)$, embedded in E_8 through $F_4 \times G_2 \subset E_8$, is a conjugate of the maximal subgroup ${}^3D_4(2){:}3$ of $\mathrm{Th} \subset E_8(\mathbb{F}_3)$. It is not. Under $F_4 \times G_2$ the adjoint representation is $248 = (52,1) \oplus (1,14) \oplus (26,7)$,

and any homomorphism ${}^3D_4(2){:}3 \to G_2$ factors through the quotient of order 3; so our copy of ${}^3D_4(2){:}3$ acts on $\mathfrak{e}_8$ by $\chi_{52} + (14 \text{ characters of order dividing } 3) + \chi_{26} \otimes (7 \text{ such characters})$. The restriction of the 248-dimensional character of Th to its maximal subgroup is $\chi_{52} + \chi_{196}$ (computed with the character table library of GAP), which has no such form. The comparison that matters is modulo 3, where the characters of order 3 become trivial and $196 = 14 + 7 \cdot 26$ in degree; but the Brauer characters still differ (on involutions, 24 for ours and -8 for Th's), and the 3-modular composition factors are $52 + 1^{21} + 25^7$ for ours and $52 + 196$ for Th's. The two copies of ${}^3D_4(2){:}3$ in $E_8(\mathbb{F}_3)$ are therefore different: the thread to the Monster runs through isomorphism type, not through the embedding in E_8.

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